Maternal Health Risk Stratification – v4

Label-noise aware · Still 3-class · Pipeline + class_weight + GridSearch

Key result: After removing true label conflicts (identical features, different RiskLevel) + exact duplicates, High-risk Recall reached **0.95** and Accuracy **80.3%** while remaining a 3-class problem (Low / Mid / High). Educational only – not a diagnostic tool.

What changed vs previous versions

- Explicit removal of **215 label-conflict rows** (35 feature groups with identical features but different RiskLevel)
- Exact duplicates still removed (standard practice to avoid leakage)
- HeartRate < 40 removed (2 records)
- Final clean size: **380** high-quality records (was 451)
- class_weight='balanced' + limited GridSearchCV (scoring=recall_macro) + full Pipeline
- Still 3-class – did **not** convert to binary High vs Rest

Q1 – Population & cleaning

Raw 1014 rows → HeartRate anomalies (2) + true label conflicts (215) + exact duplicates removed → **380 unique clean records**. RiskLevel as ordered category. Low 53.4%, High 26.6%, Mid 20.0%.

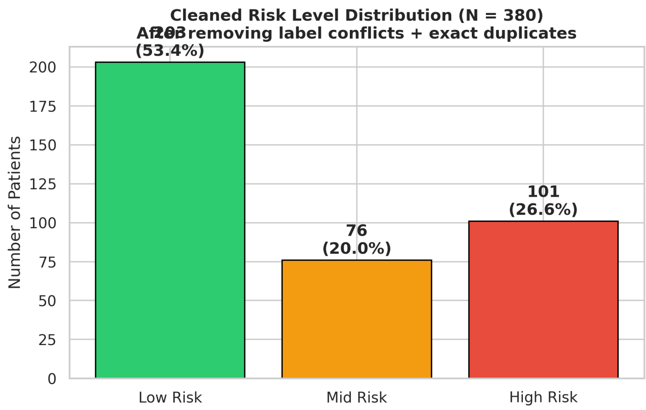

Figure 1. Cleaned risk distribution (N = 380).

Q2 – Feature differences

BS and SystolicBP remain the strongest separators. High-risk median BS is clearly elevated compared with Low/Mid groups.

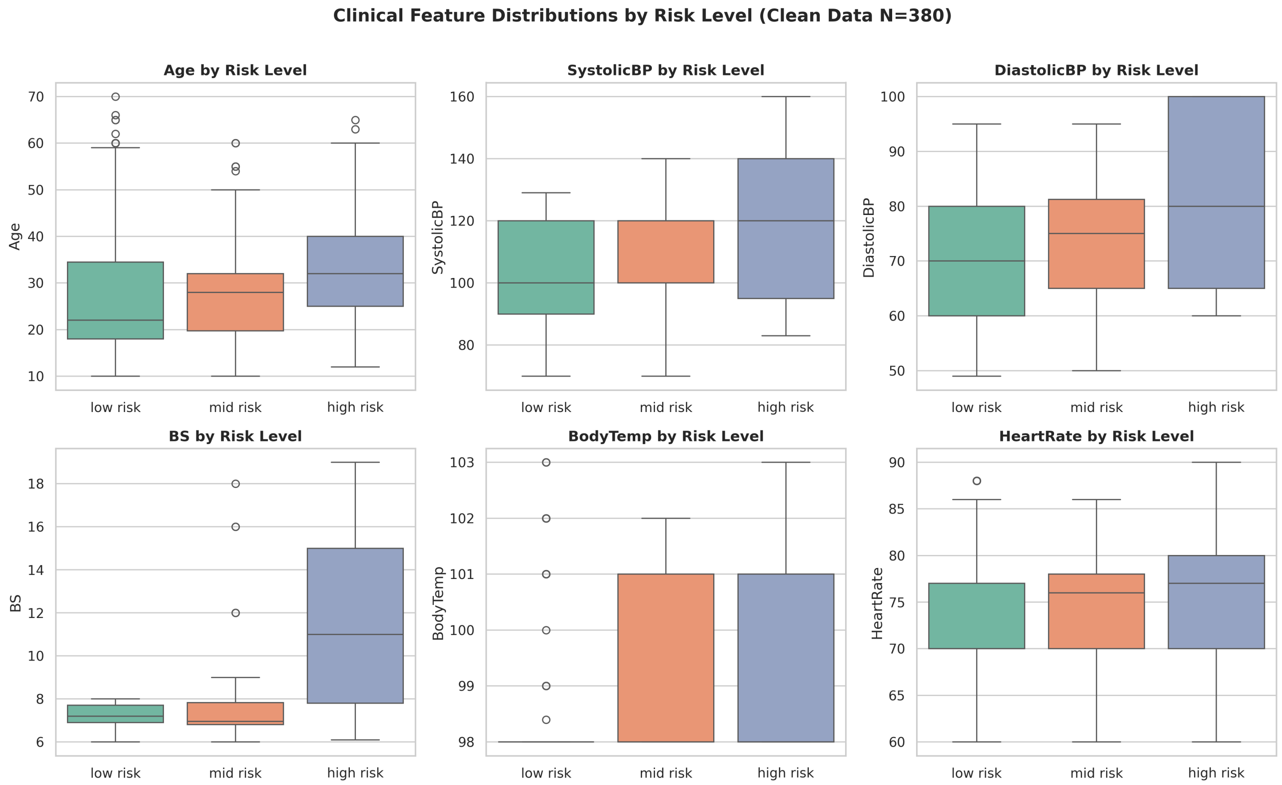

Figure 2. Box-plots by risk level.

Q3 – Association with High Risk

Pearson with High-Risk indicator: BS 0.61, SystolicBP 0.31, DiastolicBP 0.27, BodyTemp 0.23, HeartRate 0.19, Age 0.17. Association ≠ causation.

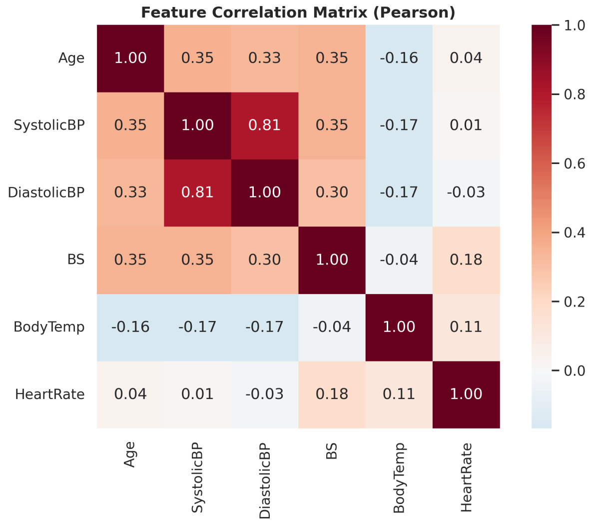

Figure 3. Correlation heatmap.

Q4 & Q5 – Models & metrics

Decision Tree (tuned, balanced): Accuracy **80.3%**, High-risk Recall **0.95**, High-risk F1 **0.95**.

Logistic Regression (balanced): Accuracy 71.1%, High-risk Recall 0.75.

High-risk Recall improved substantially (previous ~0.61 → 0.78 → **0.95**) after removing label noise and using balanced weights. Mid-risk remains the hardest class due to feature overlap.

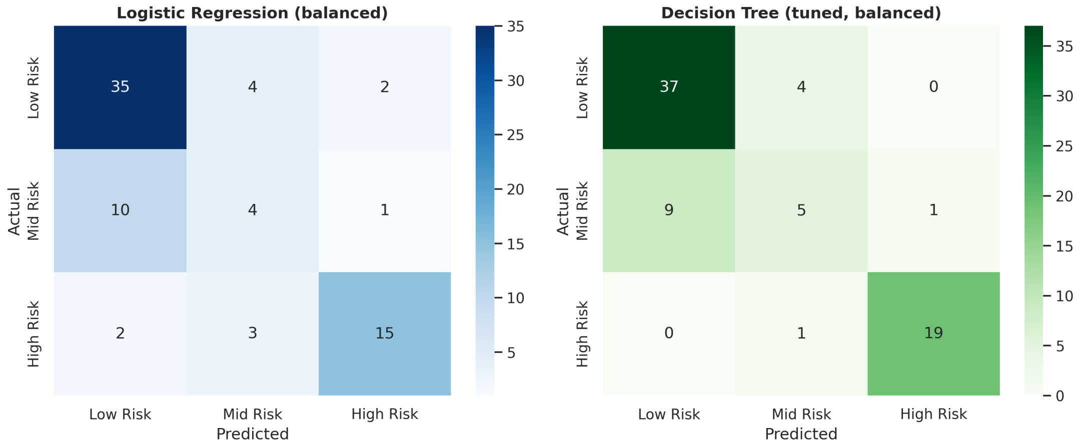

Figure 4. Confusion matrices.

Q6 – Errors

Dominant residual error is now mainly **Mid** → **Low**. High-risk cases are rarely missed (only 1 error on test set). This is a clear improvement over earlier versions where High → Low was frequent.

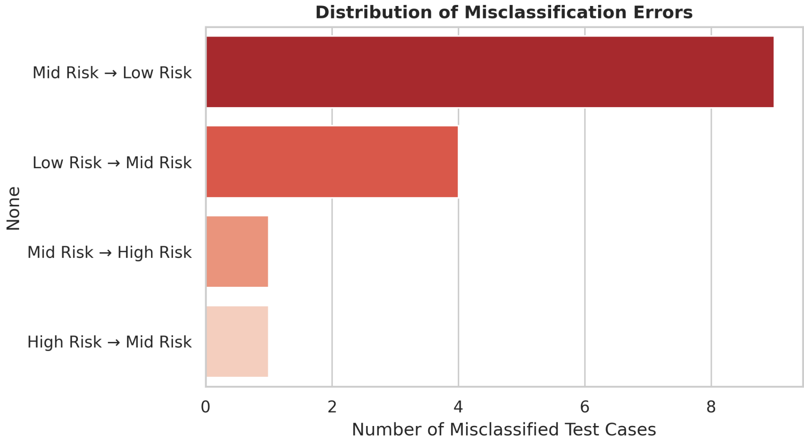

Figure 5. Error pairs.

Q7 – Drivers

DT Gini importance: BS ~41%, SystolicBP ~32%, BodyTemp ~14%, Age ~7%. LR coefficients for High-risk also rank BS highest (strong positive).

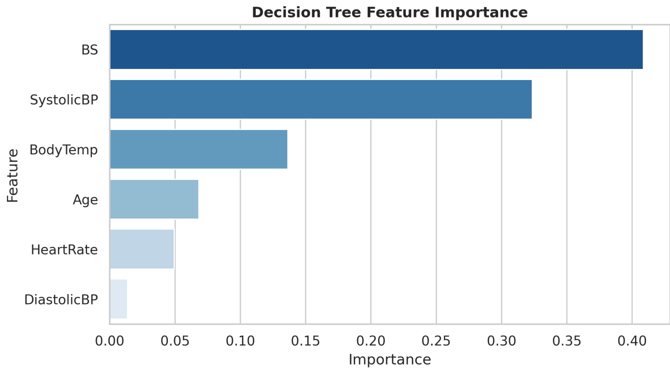

Figure 6. DT feature importance.

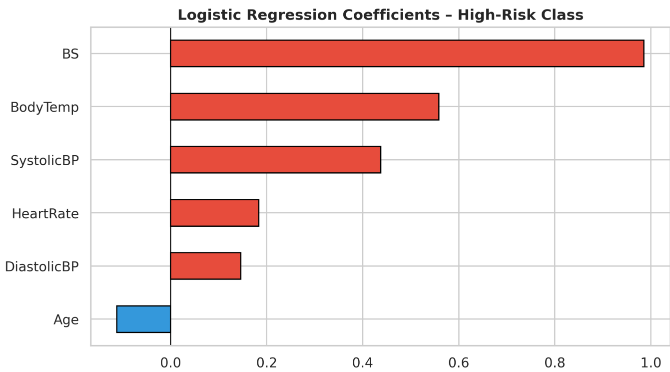

Figure 7. LR coefficients (High-risk class).

Q8 – Take-aways

1. BS and SystolicBP are the primary biomarkers in this dataset.
2. After removing true label conflicts + exact duplicates + balanced weights, High-risk Recall reached **0.95** while staying 3-class.
3. Model is still only a preliminary screening aid under clinician supervision – **not** a diagnostic tool.
4. Limitations: aggressive cleaning reduced N to 380, single geography, missing covariates (proteinuria, gestational age, BMI).
5. We deliberately did not switch to binary classification, which would look better on paper but would not answer the project questions about Mid risk.
6. Peer feedback on conflicting labels (same features, different RiskLevel) was addressed by explicit removal of those 215 rows.

Cleaning: HeartRate anomalies (2) + label conflicts (215) + exact duplicates → N=380. Best DT: criterion=entropy, max_depth=4, min_samples_leaf=3. random_state=42.